

Programming Assignment #1 : Differential Evolution

1. Refer to the Wikipedia page:

https://en.wikipedia.org/w/index.php?title=Test_functions_for_optimization&oldid=1020702629

(The relevant sections of that page are copied in this assignment.)

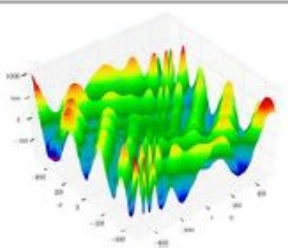
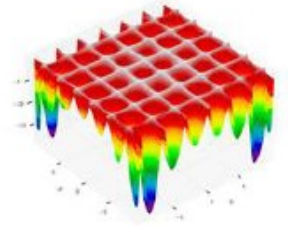
You need to develop your own program, in Python language, that performs DE optimization. The step-by-step process for this is given in the lecture slides, along with the logic and formulae for generating the Mutant vector, Trial Vector, and for performing Selection.

For generating the Mutant Vector, use the formula:

$$V_{i,g} = X_{r1,g} + R \times (X_{best,g} - X_{r1,g}) + F \times (X_{r2,g} - X_{r3,g})$$

The above formula is similar to the one given in the slides (where $X_{best,g}$ denotes the best candidate in the previous generation). Take R as 0.5 and F varies randomly between [-2, 2] in a generation but is same for all candidates in the generation.

Use this program to find the optima for two of the functions in that Wikipedia page – the *Eggholder function*, and the *Holder Table function*. The relevant section of the page is reproduced below:

Function Name	3-D Plot	Objective function – Note only 2 paras - (x, y) involved
Eggholder function		$f(x, y) = -(y + 47) \sin \sqrt{\left \frac{x}{2} + (y + 47) \right } - x \sin \sqrt{ x - (y + 47) }$
Hölder table function		$f(x, y) = - \left \sin x \cos y \exp \left(1 - \frac{\sqrt{x^2 + y^2}}{\pi} \right) \right $

The range of parameters for the Eggholder function is: $-512 \leq x, y \leq 512$, while for the Holder Table function is $-10 \leq x, y \leq 10$. There are no explicit constraints in these problems.

The same DE program should work on both, as well as any other single-objective optimization problem you would like to solve.

Your optimized values should match the values given in the link page. You will need to make plots of the convergence history of the average (across all candidates) and the best (candidate) value of the objective function across all generations (gens. on x-axis). [Note that

the identity of “best candidate” will change across generations; the plot should track the best value independent of candidate]. Both these parameters should appear on the same plot; however, the two functions are to be plotted on two different plots. You need to experiment with two parameters – the population size and the max. number of generations for solution. Do you find similar trends for both functions?

Ball-park figures for parameters:

1. Population size: 20, 50, 100, 200.
2. Num of Gens: 50, 200.
3. Crossover Probability : 0.80
4. Constants R and F for creating mutant vector: R = 0.5; for F, choose between -2 and +2 randomly across generations, but keep it same for all candidates within a generation.

Your submission should be a folder containing your code, a word doc containing comparison of your optimized solution against the correct solution, as well as the plots for the given cases. Ideally all comparison cases, or at least some of them.

BONUS: 20%. Write the code for PSO for optimization of the same two problems. In place of crossover probability, use $c1 + c2 = 4$. Choose your own values of $c1$ and $c2$. Plot the same set of convergence characteristics (as you have done for DE), for PSO. Put the PSO results also in your final submission folder.